

Pulmi Wijesinghe

DATA SCIENCE UNDERGRADUATE | ASPIRING DATA ENGINEER

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Portfolio: <https://pulmivihansa-portfolio.vercel.app/>

Professional Summary

Data Science undergraduate at Sri Lanka Institute of Information Technology (SLIIT) with hands-on experience in Machine Learning, Data Analytics, Data Warehousing, ETL pipeline development, and Full-Stack Development through academic and team projects. Skilled in Python, SQL, Power BI, SSIS, React.js, Node.js, and MongoDB, with practical experience in data preprocessing, data transformation, statistical analysis, dashboard development, and RESTful API development. Experienced in developing data-driven applications, analytical solutions, and scalable backend systems using modern software engineering practices. Strong understanding of AI/ML concepts, data visualization, and business intelligence workflows. Passionate about leveraging data-driven technologies to solve real-world problems and further developing expertise in data engineering and analytics.

Technical Skills

- **Programming Languages:** Python, Java, SQL, JavaScript, TypeScript, C#, C++
- **Data Engineering & Analytics:** ETL, SSIS, Data Warehousing, Data Modeling, SCD Type 2, SSAS
- **Analytics & BI:** Power BI, Dashboard Development, Statistical Analysis, Machine Learning
- **Databases:** MySQL, MongoDB, Oracle, SQLite
- **Web & Mobile Development:** HTML, CSS, Tailwind CSS, React.js, Node.js, Express.js, React Native, MERN Stack, Kotlin
- **Frameworks & Libraries:** Vite, Three.js
- **Tools & Technologies:** Git, GitHub, VS Code, Visual Studio, Android Studio, Postman, SQL Workbench, RStudio
- **Development Practices:** RESTful API Development, MVC Architecture, Agile Methodology

Projects Experience

Data Warehouse & BI Analytics Project | ETL Pipeline, OLAP & Dashboard Development | Feb 2026 – Apr 2026

SQL Server, SSIS, SSAS, Power BI

- Architected a star schema data warehouse with 1 fact table and multiple dimension tables for analytical reporting.
- Engineered ETL pipelines using SSIS to extract, transform, and load data from CSV and TXT data sources into staging and warehouse layers.
- Utilized data transformations, lookup operations, surrogate key generation, and accumulating fact table processing within ETL workflows.
- Constructed SSAS multidimensional cubes supporting OLAP operations including roll-up, drill-down, slice, and dice analysis.
- Produced interactive Power BI dashboards featuring drill-through reports, cascading slicers, KPI cards, and trend visualizations.
- Leveraged dimensional modeling, data warehousing, and business intelligence concepts to support scalable analytics workflows.

Smart Campus Operations Hub | Role Management, OAuth Integration Enhancements & Notification | Feb 2026 – Apr 2026

Spring Boot, React.js, MongoDB, OAuth 2.0

- Developed and integrated RESTful APIs for user authentication, role management, approval workflows, and notification processing using Spring Boot.
- Configured secure Google OAuth 2.0 login and JWT-based authentication with QR code-based Two-Factor Authentication (2FA) using Google Authenticator.
- Architected Role-Based Access Control (RBAC) for Admin, Student, and Technician roles to enforce secure system access.
- Managed user registration, login, signup approval, and user management workflows with validation and secure access handling.

- Established a real-time notification system supporting 3 user roles with prioritized alerts, read/unread management, and admin notification controls.
- Designed responsive React.js interfaces for authentication, user approval dashboards, RBAC management, and notification management workflows.
- Collaborated within a 4-member Agile team using GitHub version control, branch management, and module integration practices.

Machine Learning Analytics Project | Noise Impact on Model Robustness | Feb 2026 – Apr 2026

Python, scikit-learn, React, Vite

- Performed an end-to-end machine learning analytics project using the Red Wine Quality dataset (1,599 records, 11 features).
- Simulated 3 data quality issues: Gaussian noise, missing values, and outliers across multiple test scenarios.
- Trained and compared 3 regression models: Linear Regression, Random Forest, and Support Vector Regressor (SVR).
- Assessed model robustness using RMSE, MAE, and R^2 under noisy conditions.
- Conducted statistical analysis using correlation profiling and one-way ANOVA to validate performance differences.
- Developed an interactive React dashboard with dynamic charts and filters to visualize model degradation trends.
- Automated analytics reporting workflows using Python pipelines for frontend-ready result generation.

PackPal – Smart Bag Selling System | Full-Stack Development | Aug 2025 – Oct 2025

MERN Stack, Postman

- Developed a full-stack e-commerce platform for smart bag sales with inventory management features including stock tracking, product updates, low-stock handling, and real-time inventory monitoring.
- Implemented secure authentication, order processing, and finance management features to improve system reliability and user experience.

Personal Portfolio Website | Front-End Development | Dec 2025

TypeScript, React.js, Git/GitHub, Vercel

- Created a responsive personal portfolio website to showcase projects, achievements, and technical skills.
- Optimized responsive design and cross-device compatibility to improve accessibility and user experience across desktop and mobile platforms.

CrownCrest – Online Hotel Reservation System | Full-Stack Development | Feb 2025 – Apr 2025

Java, JSP/Servlet, MVC, Apache Tomcat, HTML, CSS, JavaScript

- Engineered a hotel reservation platform supporting real-time room availability checking, online reservations, and secure authentication workflows.
- Enabled secure user authentication and admin management functionalities using the MVC architecture pattern.

Education

Sri Lanka Institute of Information Technology (SLIIT) | Sep 2023 – Present | Expected Graduation: 2027

BSc (Hons) Information Technology specializing in Data Science (Undergraduate)

Sanghamiththa Balika Vidyalaya, Galle | 2008 – 2021

GCE Advanced Level (Physical Science Stream)

Certifications

MongoDB Python Developer Path – MongoDB Academy

MongoDB Data Modeling Path – MongoDB Academy

AI/ML Engineer – Stage 1 by SLIIT

AI/ML Engineer – Stage 2 by SLIIT

Python Programming - University of Moratuwa

Introduction to Career Skills in Data Analytics – LinkedIn Learning